Appendix A. Retrieval Benchmark Comparison

This appendix compares the three vector-store options used in Track B: Chroma, FAISS, and Qdrant.

The benchmark corpus is generated at three sizes: 6, 100, and 1000 documents. The measurements below come from the actual benchmark runs recorded in results.csv.

A.1 Quantitative Benchmark Matrix

Store	Documents	Index Time (s)	Avg Latency (ms)	Query Count	Notes
Chroma	6	0.120899	1.920	3	Measured run
Chroma	100	0.154131	1.978	3	Measured run
Chroma	1000	0.413993	2.439	3	Measured run
FAISS	6	0.000084	0.069	3	Measured run
FAISS	100	0.000133	0.066	3	Measured run
FAISS	1000	0.000578	0.203	3	Measured run
Qdrant	6	0.012289	0.828	3	Measured run
Qdrant	100	0.036456	1.148	3	Measured run
Qdrant	1000	0.263583	3.485	3	Measured run

A.2 Qualitative Comparison

Store	Install Effort	Persists to Disk	Metadata Filtering	Practical Notes
Chroma	Low to moderate	Yes, with a persistent client and persistence directory	Yes	Easy to use for local prototyping and supports metadata-based patient lookup.
FAISS	Low	Not by default as a full vector database	No native metadata filtering	Fastest raw search path, but you need a separate metadata store for patient_id filtering and persistence.
Qdrant	Moderate	Yes, when run with storage enabled	Yes	Strong choice when you need both vector search and payload filtering in one system.

A.3 Interpretation

The measured runs show the same overall pattern across all three systems:

- FAISS has the smallest index creation time and the lowest query latency at all three scales.
- Qdrant remains fast enough for interactive retrieval and gives native metadata filtering.

- Chroma is easy to use for local experimentation, but its build/query overhead is higher than FAISS.

For Track A style patient-specific retrieval, metadata filtering matters because queries may need to restrict results by `patient_id`. On that criterion, Chroma and Qdrant are more suitable than raw FAISS unless you add a separate metadata layer.

If the goal is pure speed, FAISS wins. If the goal is practical retrieval with filtering and persistence, Qdrant is the most balanced option. Chroma is a reasonable middle ground for rapid development and demonstrations.